Animal Form and Function II

Short Notes

By

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Protonephridial excretory system:

The protonephridial excretory system is composed of a network of excretory canals that open to the outside of the body through excretory pores.

Flame cell:

A flame cell is a specialized excretory cell found in the simplest freshwater invertebrates, including flatworms, rotifers and nemertean's (ribbon worms); these are the simplest animals to have such an excretory system.

Metanephridium:

Metanephridium is a type of excretory gland found in invertebrates such as annelids, arthropod and Mollusca. Unlike protonephridia, the metanephridia open internally to the body fluids and are multicellular.

Nephridiopore:

In most annelids excretory system Each tubule leads to an enlarged bladder that empties to the outside of the body through an opening called the nephridiopore

Antennal gland:

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The antennal gland, lies in front of and on both sides of the esophagus. The gland is divided into An End Sac, where fluid collects by filtration, And A Labyrinth, the walls of which are greatly folded and is glandular and an important site for reabsorption.

Malpighian tubules:

In the insects the Malpighian tubules are present which remove nitrogenous wastes (uric acid) from the hemocoel. Some water, ions, and organic compounds are reabsorbed in the basal portion of the Malpighian tubules and the hindgut.

Coxal glands:

In arachnids the coxal glands are present to collecting and excreting urine. They are present on one to several pairs of appendages near the proximal joint (coxa) of the leg.

Gaining and losses process of water in vertebrates:

Vertebrates gain water from liquids and solid foods, and They lose water by evaporation.

Filtration:

In this process blood passes through a filter that retains blood cells, proteins, and other large solutes but lets small molecules, ions, and urea pass through.

Reabsorption:

Reabsorption is a highly selective process that returns usable materials to the blood but leaves wastes and excesses of other substances to be excreted in the urine.

Secretion:

In this process select ions and end products of metabolism (e.g., K, H, NH_3) that are in the blood are added to the filtrate for removal from the body.

Renal capsule:

Each kidney has a coat of connective tissue called the renal capsule.

Medulla:

The inner portion of the kidney is called the medulla.

Cortex:

the region between the capsule and the medulla is the cortex.

Kinds of kidneys:

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Pronephros, Mesonephros and Metanephros.

Pronephros:

In vertebrates such as in hagfishes, pronephros remains as the functioning kidney. The primitive pronephric kidney is found in adult hagfishes and in the embryos of fishes and amphibians.

Mesonephros:

It is a paired organ consisting of a set of nephrons having capsules that filter blood from the glomerulus and tubules whose cells reabsorb water and nutrients and secrete nitrogenous wastes. It serves as the main excretory organ of aquatic vertebrates and as a temporary kidney in reptiles, birds, and mammals.

Mesonephros:

The mesonephros is the functional embryonic kidney of many vertebrates and also adult fishes and amphibians.

Excretion in Sharks:

Sharks and their relatives (skates and rays) have mesonephric kidneys and have solved their osmotic problem in ways different from the bony fishes. Instead of actively pumping ions out of their bodies through the kidneys, they have a rectal gland that secretes a highly concentrated salt (NaCl) solution.

How reduce water loss in sharks:

To reduce water loss, they use two organic molecules—urea and trimethylamine oxide (TMO)—in their body fluids to raise the osmotic pressure to a level equal to or higher than that of the seawater.

Excretion in Teleost Fishes:

Most teleost fishes have mesonephric kidneys. Since the body fluids of freshwater fishes are hyper-osmotic relative to freshwater, water tends to enter the fishes, causing excessive hydration or bloating. They absorb salts and ions by active transport across their gills. They filter large quantities of water from the blood and tubules reabsorb some ions from the filtrate. , Ca2, Mg2, SO4 and PO4 • out of the extracellular fluid and into the nephron tubes. The ions are then excreted in a concentrated urine.

Excretion in Amphibians:

The amphibian kidney is identical to that of freshwater fishes. Amphibians take up water and ions in their food and drink, through the skin that is in contact with moist substrates, and through the urinary bladder. The kidney forms a dilute urine by reabsorbing Na and Cl ions. When water leaves the urinary bladder, the remaining ions are reabsorbed.

Reptiles Birds and Mammals:

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In most reptiles, birds, and mammals, the kidneys can remove far more water than can those in amphibians, and the kidneys are the primary regulatory organs for controlling the osmotic balance of the body fluids.

How to reduce mammal's evaporative loss:

To reduce evaporative loss, many mammals have nasal cavities that act as countercurrent exchange systems.

why a dog's nose is usually cold and moist:

As the air cools, much of the water condenses on the nasal surfaces. so dog's nose is usually cold and moist.

Excretory system:

The excretory system is a biological system that removes excess, unnecessary materials from the body fluids of an organism, so as to help maintain internal chemical homeostasis and prevent damage to the body.

What is kidney:

The kidney is a bean-shaped structure with a convex and a concave border. A recessed area on the concave border is the renal hilum, where the renal artery enters the kidney and the renal vein and ureter leave.

What is the urinary bladder:

The urinary bladder is the organ that collects waste excreted by the kidneys prior to disposal by urination. It is a hollow muscular, and distensible (or elastic) organ, present on the pelvic floor.

Renal pyramid:

The kidney is divided into two major structures: The outer renal cortex and the inner renal medulla. these structures take the shape of 85 to 18 cone-shaped renal lobes, each containing renal cortex surrounding a portion of medulla called a renal pyramid.

Medulla:

A frontal section through the kidney reveals an outer region called the renal cortex and an inner region called the medulla.

Renal papillae:

The papillae are bundles of collecting ducts that transport urine made by nephrons to the calyces of the kidney for excretion.

Nephron:

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The basic structural unit of kidney is called nephron. There are more than 1000000 nephrons in each kidney which filters the blood and produce urine.

Bowman's capsule:

The cup-shaped corpuscle containing the glomerulus is called the Bowman's capsule.

Structure and function of the epithelial cells:

The structure and function of the epithelial cells lining the lumen of the nephron change during the course of the nephron, and have segments named by their location which reflects three different functions: • 1. Filtration • 2. Reabsorption • 3. Secretion

Why Water cannot flow out of the ascending limb:

Water cannot flow out of the ascending limb because the cells of the ascending limb are impermeable to water.

In which thing the production of concentrated urine depends:

Production of concentrated urine depends on a high salt and urea concentration in the interstitial fluid of the kidney medulla.

What is urine:

Urine is a watery solution of nitrogenous wastes, excess salts, and other substances not needed by the body.

How these Vertebrates Maintain Water and Salt Balance:

Freshwater Fishes:

Environmental Concentration relative to body fluids is hypoosmotic. • Urine concentration relative to blood is hypoosmotic. • Major nitrogenous waste is ammonia. • Key adaptation: absorb ions through gills.

Seawater Fishes:

Environmental Concentration relative to body fluids is hyperosmotic. • Urine concentration relative to blood is isoosmotic. • Major nitrogenous waste is ammonia. • Key adaptation: secrete ions through gills.

Sharks:

Environmental Concentration relative to body fluids is isoosmotic. • Urine concentration relative to blood is isoosmotic. • Major nitrogenous waste is ammonia. • Key adaptation: secrete ions through rectal glands..

Amphibians:

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Environmental Concentration relative to body fluids is hypoosmotic. • Urine concentration relative to blood is very hypoosmotic. • Major nitrogenous waste is ammonia and urea. • Key adaptation: absorb ions through skin.

Marine reptiles:

Environmental Concentration relative to body fluids is hyperosmotic. • Urine concentration relative to blood is isoosmotic. • Major nitrogenous waste is ammonia and urea. • Key adaptation: secrete ions through salt glands.

Marine Mammals:

Environmental Concentration relative to body fluids is hyperosmotic Urine concentration relative to blood is very hyperosmotic. • Major nitrogenous waste is urea. • Key adaptation: drink some water.

Desert Mammals:

Environmental Concentration relative to body fluids is non comparison. • Urine concentration relative to blood is very hyperosmotic.. • Major nitrogenous waste is ammonia and urea. • Key adaptation: produce metabolic water.

Marine reptiles:

Environmental Concentration relative to body fluids is no comparison. • Urine concentration relative to blood is weakly hyperosmotic. • Major nitrogenous waste is uric acid. • Key adaptation: drink sea water and use salt glands.

Terrestrial Birds:

Environmental Concentration relative to body fluids is no comparison. • Urine concentration relative to blood is weakly hyperosmotic. • Major nitrogenous waste is uric acid. • Key adaptation: drink fresh water.

How many hormones that regulate the homeostatic functions of the kidney:

There are many hormones that regulate the homeostatic functions of the kidney. These are: • Renin–Angiotensin–Aldosterone, Endothelin · Natriuretic Hormones, Parathyroid Hormone, Antidiuretic Hormone (ADH).

Function of ADH:

ADH makes the collecting ducts more permeable to water so that more water is reabsorbed. • As a result, a small volume of concentrated urine is produced.

Kidney failure:

Kidney failure the last stage of chronic kidney disease. Kidney failure means when they stop working well enough for survival without dialysis or a kidney transplant.

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Kidney stones:

Kidney stones are hard deposits made of minerals and salts that form inside the kidneys. Also called renal calculi.

How many Types of Kidney Stones:

Calcium Oxalate Stones. (The most common type of kidney stone.) • Calcium Phosphate Stones. • Uric Acid Stones. • Cystine Stones.

Lithotripsy:

It is a procedure that uses shock waves or LASER (Light Amplification by Stimulated Emission of Radiation) to break up stones into small particles in the kidney and parts of the urete

UTI is caused by:

UTI is caused by microorganisms — usually bacteria — that enter the urethra and bladder, causing inflammation and infection. • Though UTI most commonly happens in the urethra and bladder, bacteria can also travel up the ureters and infect the kidneys.

Polycystic kidney disease (PKD):

It is an inherited disorder in which clusters of cysts develop within the kidneys, causing kidneys to enlarge and lose function over time.

Hydronephrosis:

Hydronephrosis is a condition that typically occurs when a kidney swells due to urine failing to properly drain from the kidney to the bladder. • This swelling most commonly affects only one kidney, but it can involve both kidneys. • Hydronephrosis is not a primary disease.

Thermoregulation:

Thermoregulation is a complex and important physiological process for maintaining heat homeostasis despite environmental changes.

Reproduction:

Reproduction is the basic characteristic of living things. Reproduction is giving rise of new individuals like their parents.

Types of reproduction:

There are two types of reproduction: • 1. Asexual reproduction and • 2. Sexual reproduction.

. Asexual reproduction:

the organisms reproduce by mitotic divisions producing offspring that are identical to their parents.

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Sexual reproduction:

Involves the fusion of male and female gametes i.e sperms and eggs to form zygote. • This zygote develops to form a unique new individual having two kinds of cells: • somatic cells and • germ cells

Binary Fission:

Binary Fission is the division of one cell, body, or body part into two. • In this process, the Amoeba pinches in two by an inward furrowing of the plasma membrane. • Binary fission is common in protozoa (Amoeba, Plasmodium, Paramecium etc)

Multiple fission:

In this process, the nucleus divides mitotically to produce a large number of nuclei before the cell divides. • Each nucleus, with the surrounding cytoplasm, forms a daughter cell.

Budding:

Budding is a form of asexual reproduction that results from the outgrowth of body region leading to a separation from the original organism into two individuals. • It may be External or Internal.

Internal budding:

Internal budding is by the formation of gemmules which are collections of many cells surrounded by a body wall.

When the body of the parent dies and degenerates, each gemmule gives rise to a new individual.

Fragmentation:

Fragmentation is a type of asexual reproduction that occurs in some cnidarians, platyhelminths, and echinoderms. • For example, in sea anemones.

Parthenogenesis:

Parthenogenesis is a form of asexual reproduction in which development of egg occur without being fertilized. Types of Parthenogenesis: • Diploid Parthenogenesis • Haploid Parthenogenesis.

Diploid Parthenogenesis:

In this type, no meiosis occurs and the eggs are formed by mitosis. In some cases egg are formed by modified form of meiosis involving total nondisjunction of the chromosomes thus retaining the diploid number. • Such eggs develop into young females.

Haploid Parthenogenesis:

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In this type of Parthenogenesis haploid ova are formed by meiosis and they may or may not be fertilized by sperm. • If fertilized, the zygotes develop into females (Queen or sterile female workers in bee). • If unfertilized the eggs develop to become haploid males (drones).

Advantages and Disadvantages of Asexual Reproduction:

Advantages: Large number of offspring are produced very quickly from only one parent when conditions are favorable. Large number of organisms mean that species may survive when conditions or number o predators change. Energy is not required to fin mate.

Disadvantages: Offspring are genetic clones. 2. A negative mutation can make asexually produced organisms susceptible to disease and can destroy large number of offspring.

Fertilization:

Fertilization is the fusion of haploid gametes, egg and sperm, to form the diploid zygote. • There are delicate differences in the fertilization process which occurs naturally within the body or through reproductive technologies outside the body.

Stages of Fertilization Process:

1) sperm and egg preparation, 2) sperm-egg recognition 3) fusion of sperm and egg membranes 4) Penetration of sperm nucleus into the egg. 5) Fusion of pronuclei and activation of the zygote.

External fertilization:

It is a mode of reproduction in which a male organism's sperm fertilizes a female organism's egg outside of the female's body. external fertilization is most common in amphibians and fish.

What is spawning:

The release of eggs and sperm into the water is known as spawning.

Internal fertilization:

Internal fertilization is the union of an egg cell with a sperm during sexual reproduction inside the female body.

What is spermatophores:

In some invertebrates (e.g., arrow worms, leeches, some insects) the sperms are produced in the testes and transported via a sperm duct to a storage packet area called spermatophores.

What are the various forms of sexual reproduction:

The various forms of sexual reproduction are: • 1. Hermaphroditism • 2. Sequential hermaphroditism.

Hermaphroditism:

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Hermaphroditism, is the condition of having both male and female reproductive organs. • Hermaphroditic animals are mostly invertebrates such as worms, bryozoans, trematodes (flukes), snails, slugs, and barnacles.

Protogynous:

Protogynous animals are born female and at some point of their life span change sex to males. • Examples: Bluehead wrasses, Rana temporaria.

Protandrous:

Protandrous animals are male during their early life history and female later in the life history. • Examples: Many fish, mollusks, and crustaceans.

Advantages & Disadvantages of Sexual Reproduction:

Advantages: Offspring are genetically unique because of crossing over, Independent assortment and random fusion of gametes. • Slower rate of reproduction but faster adaptation/evolution. • Favorable when the environment is not stable. • Lower extinction rate. • Fast removal of bad mutations.

Disadvantages: Slower reproductive rate, fewer offspring are produced over longer time span. • Potential for spread of sexually transmitted diseases through population. • Energetically costly, i.e. requires a lot of ongoing energy input from the parent.

Reproduction in Fishes:

All fishes reproduce in aquatic environments. • In bony fishes, fertilization is usually external, and eggs contain only enough yolk to sustain the developing fish for a short time.

Reproduction in Amphibians:

The amphibians were the first vertebrates to invade the land. • They have not, however, become adapted to a completely terrestrial environment; their life cycle is still inseparably linked to water. • Among most amphibians, fertilization is still external, just as it is among the fishes.

Reproduction in Reptiles:

The reptiles were the first group of vertebrates to completely abandon the aquatic habitat because of adaptations that permitted sexual reproduction on land. Internal fertilization protects the gametes from drying out, freeing the animals from returning to water to breed. • Many reptiles are oviparous, and the eggs are deposited outside the body of the female.

Reproduction in Birds:

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All birds are oviparous, and the eggshells are much thicker than those of reptiles. • Thicker shells permit birds to sit on their eggs and warm them. • This brooding, or incubation, hastens embryo development. • When many young birds hatch, they are incapable of surviving by their own.

Reproduction in Mammals:

The most primitive mammals, the monotremes (e.g., the duckbilled platypus and spiny anteater), lay eggs, as do the reptiles from which they evolved. • All other mammals are viviparous.

Human Male Reproductive System:

The main function of the male reproductive system is to produce sperms and male sex hormone for fertilization of the ovum.

Three categories of male reproduction:

The first category consists of organs located outside the body produces and stores sperms. The second category are the structures located around the pelvis region of male. The third category are those used for copulation and deposition of semen in the vagina (the penis and urethra).

Tunica Albuginea:

Each testis is covered by a thick capsule called Tunica Albuginea.

Mediastinum testis:

Posteriorly the tunica albuginea is thickened and is called mediastinum testis.

Septula testis:

Thin fibrous septa, called Septula testis, radiate from the midiastinum to the capsule and divide the interior of the testis into 200 to 300 Pyramidal Lobules.

What is the testes:

Testes hang outside the body, the temperature inside the scrotum is about 34° C compared to a 37° C core temperature.

Leydig cells:

Between the seminiferous tubules are clusters of endocrine cells, called Leydig cells, that secrete the male sex hormone, the testosterone.

Rete testis and epididymis:

The seminiferous tubules merge into a network of tiny tubules called the rete testis, which merges into a coiled tube called the epididymis.

Functions of epididymis:

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The epididymis has three main functions: • 1. it stores sperms until they mature and ready to be ejaculated, • 2. it contains smooth muscle that helps propel the sperm toward the penis by peristaltic contractions, • 3. it serves as a duct system for sperm to pass from the testis to the ductus deferens.

Accessory glands:

1. Seminal vesicles • 2. Prostate gland, and • 3. Bulbourethral glands.

Seminal Vesicles:

These vesicles are paired secrete water, fructose, prostaglandins, and vitamin C. • This secretion provides an energy source for the motility of sperms.

The Prostate Gland:

It is walnut sized gland, located between the bladder and penis. It secretes water, enzymes, cholesterol, buffering salts, and phospholipids.

The Bulbourethral Glands:

The bulbourethral glands (also known as Cowper's glands) are a pair of pea shaped exocrine glands urethra. • These glands secrete a clear, alkaline fluid that lubricates the urethra to facilitate the ejaculation of semen and lubricates the penis prior to sexual intercourse.

glans penis:

The corpus spongiosum extends beyond the corpora cavernosa and becomes the expanded tip of the penis called the glans penis.

Average human ejaculation:

The average human ejaculation is about 3 to 4 ml of semen and contains 300 to 400 million sperms.

Androgens and gonadotropins:

Male sex hormones are collectively called androgens. And The hormones that travel from the brain and pituitary gland to the testes and ovaries are called gonadotropins.

Labia majora:

In female reproduction enclose and protect the other external reproductive organs.

Labia minora:

They lie just inside the labia majora, and surround the openings to the vagina.

Clitoris:

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A small, sensitive protrusion at the point where the two labia minora meet. • This is comparable to the penis in males.

Vagina:

The vagina is a canal that joins the cervix (the lower part of uterus) to the outside of the body. It also is known as the birth canal.

Uterus:

The uterus is a hollow, pear-shaped organ that is the home to a developing fetus. • The uterus is divided into two parts: the cervix, which is the lower part that opens into the vagina, and the main body of the uterus is called **corpus.**

Ovaries:

The ovaries are almond sized, oval-shaped glands that are located on either side of the uterus. • The ovaries produce eggs and hormones.

Fallopian tubes:

These are narrow tubes that are attached to the upper part of the uterus and serve as pathways for the ova (egg cells) to travel from the ovaries to the uterus. Fertilization of an egg by a sperm normally occurs in the fallopian tubes.

Germinal epithelium:

The surface of the ovary is covered by a single layer of cuboidal epithelium, called germinal epithelium.

The primary follicle:

is the first morphological stage that marks the onset of follicular. • The previously flattened cell surrounding the oocyte now form a cuboidal or columnar epithelium surrounding the oocyte.

Secondary follicle:

Small fluid-filled spaces become visible between the granulosa cells as the follicle reaches a diameter of about 400 μm. • These spaces enlarge and fuse to form the follicular antrum.

Corpus luteum:

serves as a temporary endocrine tissue and continues to secrete the female sex hormones estrogen and progesterone.

WHAT IS OOGENESIS?

Oogenesis is the formation of ovum or egg. • In human oogenesis starts in the ovaries of the fetus before birth.

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There are 5 stages of oogenesis process: (MOST IMP LONG)

1. The germinal epithelial cells divide repeatedly until many (5-7) million diploid oogonia are formed. • 2. The oogonia grow to form primary oocytes that become surrounded by a layer of follicle cells. • 3. The primary oocytes undergoes the first meiotic division to become a secondary oocyte and a first polar body.(haploid). • 4. The secondary oocyte surrounded by the follicle cells develop into the secondary follicle. 5. The final product of meiosis II is the ovum and the second polar body.

Cervix:

The uterus terminates in a narrow portion called the cervix, which joins the uterus to the vagina.

The Mammary Glands:

MG are modified sweat glands. • MS produce and secrete milk. • They contain varying amounts of adipose tissue.

Reproductive Cycle:

Reproductive Cycle is a set of physiologic changes that recur regularly in the reproductive organs of female mammals.

Estrous Cycle:

: In animals with estrous cycles, the female is sexually receptive ("in heat") only during the estrus phase of the cycle. For example, In cow the estrus phase lasts avg. 24 hours in 21 days cycle.

Menstrual Cycle:

Adult female mammals without estrous cycle have the MC. • In these mammals uterus sheds the thickened lining of the endometrium along with blood and mucosal tissue through the vaginal opening if conception does not occur. In adult human females the MC is of 28 days and the menstrual period averages 5 days.

The increasing level of LH and the decreasing level of FSH bring about the release of oocyte from the Graafian follicle (ovulation).

Menopause:

When a female reaches 45 to 55 years of age, the ovaries lose their sensitivity to FSH and LH, • They stop making normal amounts of estrogen and progesterone and the monthly cycle ceases to occur permanently.

Follicle stimulating hormone (FSH) and luteinizing hormone (LH):

FSH stimulates follicle development and stimulates the granulosa cells to produce estrogen.

LH stimulates theca cells to multiply and produce androgens, which are converted to estrogen.

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Pregnancy/Gestation:

Pregnancy/Gestation, is the time during which one or more offspring develops inside a woman. • Pregnancy sets into motion a new series of physiological events. • In pregnancy ovaries are directly affected.

Oxytocin stimulate:

Oxytocin stimulate the uterine contractions during childbirth. • It is also important for lactation, and certain aspects of human behavior.

Prenatal period:

During the prenatal period, the developing individual takes its start becomes a ball of cells called a morula, and eventually becomes a blastocyst that implants in the endometrium.

Trimesters:

Pregnancy is arbitrarily divided into three-month periods called trimesters

The First Trimester:

After fertilization, usually in the upper third of the uterine tube, he zygote goes through several cleavages as it moves down the tube.

The Second Trimester:

In the second trimester (4th to 6th month), fetal growth is spectacular. • By now, the pregnant mother is aware of fetal movements. • The heartbeat can be heard with a stethoscope.

The third trimester:

The third trimester extends from the seventh month until birth. • During this time, the fetus has developed sufficiently (with respect to the circulatory and respiratory systems) to potentially survive if born prematurely. During the last month, fetal weight doubles.

The placenta:

The placenta is the organ that sustains the embryo and fetus throughout the pregnancy and through which gases are exchanged, nutrients are supplied and wastes are removed from the fetal systems to the maternal end.

Parts of placenta:

It has two components • Fetal part, develops from the chorionic sac. • Maternal part, derived from endometrium.

What is parturition:

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The birth process is called parturition. • During parturition, the mother's uterine muscles begin to contract, and the cervix begins to dilate, or open.

Lactation:

Lactation includes both milk secretion (production) by the mammary glands and milk release from the breasts. • During pregnancy, the breasts enlarge in response to increasing levels of the hormone prolactin. • Before birth, placental secretions of estrogen and progesterone inhibit milk secretion from the breasts

Fertilization:

Fertilization is the fusion of sperm and egg. • Fertilization and the subsequent establishment of pregnancy together are referred to as conception. • After ejaculation into the female reproductive tract, sperm remain alive and retain their ability to fertilize an ovum for an estimated 48 to 72 hours.

Fertilization involves four events: • i) the sperm contacts the egg and recognition occurs. • ii) the sperm or sperm nucleus enters the egg. • (iii) the egg becomes activated, and certain developmental changes begin. • (iv) the sperm and egg nuclei fuse.

The zygote:

The zygote formed as a result of fertilization is totipotent, i.e. it gives rise to all the cell types of the new individual. The zygote initially divides to form a 2, to 4 , 4 to 8 celled embryo. Egg Types.

On the basis of the amount of yolk how many types of eggs:

On the basis of the amount of yolk, eggs are grouped into four types: • 1. Alecithal Egg: When the egg contains no yolk. Example eutherian mammals. • 2. Microlecithal Egg: When the egg contains small or negligible amount of yolk. Example, Amphioxus, Tunicates • 3. Mesolecithal egg: When the amount of yolk present is moderate and is not high. Example, Dipnoi, Petromyzon, and amphibians. • 4. Macrolecithal Egg: when the egg contains large amount of yolk. Such eggs are also called megalecithal or Polylecithal Example, Reptiles, Birds, Prototheria (Monotremata) Egg laying mammals.

Echinoderm Embryology:

Echinoderms (Sea Urchin) exhibit radial holoblastic cleavage. • The first and second clevages are both meridional and at right angles to each other. • It means the cleavage furrow pass through the animal and vegetal poles. • The third cleavage is at right angles to the first two, dividing the embryo into 8 cells.

Amphibian Cleavage:

Cleavage in amphibians is radially symmetrical and holoblastic. • The amphibian egg, contains much more yolk which is concentrated in the vegetal hemisphere. • Here yolk is an obstruction to cleavage.

Avian Embryology:

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Fertilization of the chick egg occurs in the oviduct, before the albumen and the shell are secreted upon it. • The egg is telolecithal with a small disc of cytoplasm sitting atop a large yolk. • Cleavage occurs only in the blastodisc, a small disc of cytoplasm 2-3 mm in diameter at the animal pole of the egg.

Primitive Streak, Primitive groove, Primitive pit:

The major structural characteristic of avian, gastrulation is the **primitive streak.**

As cells converge to form the primitive streak, a depression forms within the streak. This depression is called the **primitive groove**.

The center of Hensen's node contains a funnel-shaped depression, called **the primitive pit.**

Hensen's node:

At the anterior end of the primitive streak is a regional thickening of cells called Hensen's node

Gastrula:

three layered embryo is called gastrula which has formed undergoing a process, the Gastrulation.

Gastrulation:

It is the process of highly integrated cell and tissue movements and their rearrangements so as to develop a three layered embryo composed of ectoderm, mesoderm and endoderm.

Ectoderm:

The ectoderm develops from the embryo's epiblast. • The ectoderm develops into the surface ectoderm, neural crest, and the neural tube.

The endoderm:

The endoderm is one of the germ layers formed during animal embryonic development. • Cells migrating inward along the archenteron form the inner layer of the gastrula, which develops into th endoderm.

The mesoderm:

The mesoderm forms in the embryos of triploblastic animals. • During gastrulation, some of the cells migrating inward contribute to the mesoderm, an additional layer between the endoderm and the ectoderm. • The formation of a mesoderm leads to the development of a coelom.

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